

Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

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Problem

Unsafe Long-Distance Bus Transportation in Sri Lanka

- Drivers become tired while driving
- Breaking traffic rules
- hazardous overcrowding and footboard riding.
- No real-time monitoring system
- Slow emergency response

Motivation

- Make bus travel safer
- Reduce accidents
- Eliminate accidents like bus roll over due to overcrowding and speed driving
- Improve emergency response



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Existing Systems

- Tracking ,Passenger and fatigue monitors are isolated and reactive.
- No roll-over prediction warning using passenger count and speed
- No real time road rule violation system for public transportation in Sri Lanka.

The Research Gap

- Integrated system combining passenger counting, footboard detection, and predictive rollover risk using real-time occupancy.
- Real time road rule violation detection system trained with primarily collected Local Dataset.



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Solution Overview

A smart real-time bus monitoring and safety system for long-distance buses in Sri Lanka.

Key Features :

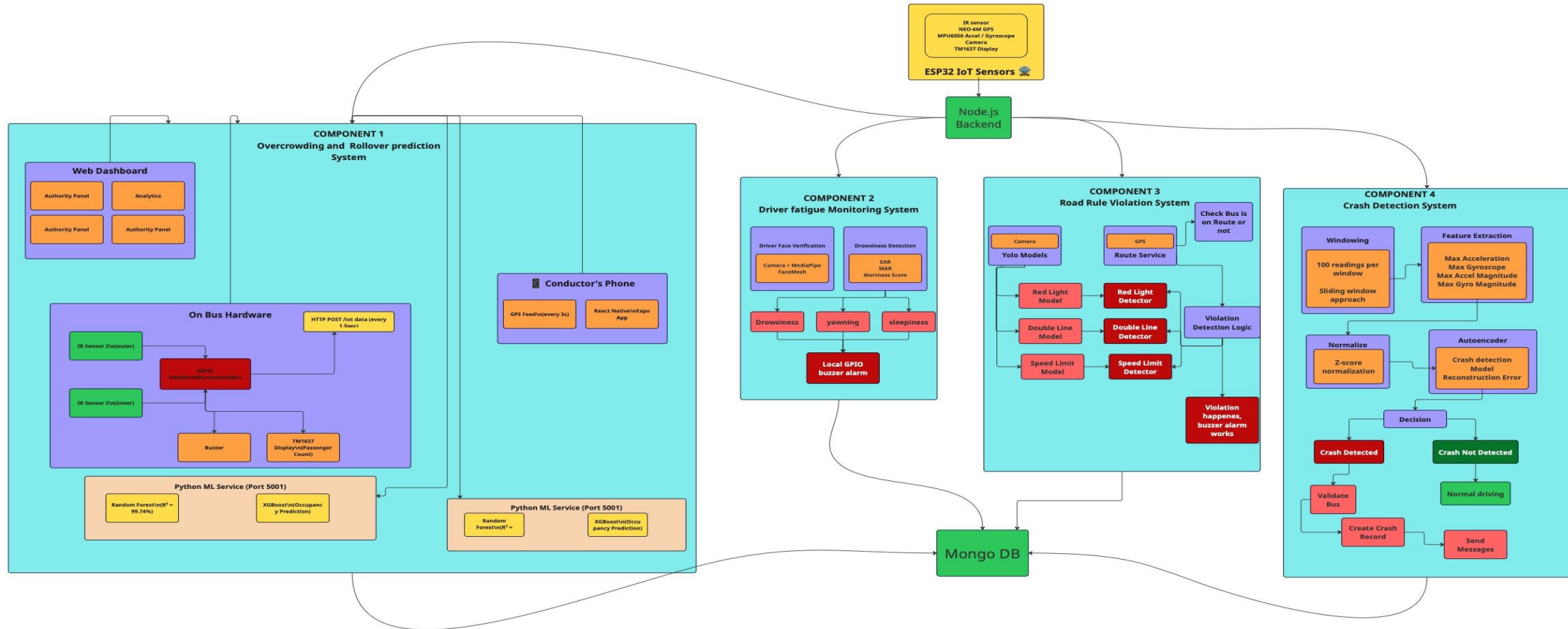
- Driver fatigue monitoring
- Passenger overcrowding detection
- Driver warning system – Roll over prediction , Distant to stop
- Real time Traffic violation detection
- Emergency crash detection and response

Non-Functional Focus:

- Real-time performance
- System reliability
- Secure data management
- User-friendly interface



System Architecture and Integration



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System Maturity & Core Innovations

- Multi-Output RandomForest handling correlated outputs (Risk Score & Stopping Distance).
- OSMnx road lookahead will detect road ahead 120m.
- Backend Optimizations – Road ahead Caching , Weather Caching improves speed
- This AI system designed from the ground up for Sri Lankan public transport, not adapted from Western models, but built for our roads, our traffic behavior, and our enforcement needs.
- Dual Threshold Anomaly Detection – ML based, Physical Impact verification.
- Autoencoder Approach – Trained on Normal Driving Data

Evaluation Strategy & Evidence-Based Results

- ML Model achieves $R^2=99.74\%$ and $MSE=0.12$, validated against physics equations.
- Mobile parallel fetching reduced cycle time from 4s → 200m
- YOLO model achieved 99.5% mAP50 on 230 test images with 4.3ms inference - proven through rigorous evaluation on diverse data.
- Normal driving test and simulated crash test. For normal driving, acceleration is 0.3 m/s^2 and reconstruction error is 0.03–0.05. For a crash event, acceleration $> 15 \text{ m/s}^2$ and reconstruction error $>$ than 0.08.



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Commercialization

Target Users / Market:

- Long-distance bus operators in Sri Lanka
- Transport authorities
- Bus passengers

Value Proposition:

- Improves passenger safety
- Reduces accidents through real-time monitoring
- More effective than manual supervision

Cost & Feasibility:

- Low-cost IoT and camera-based system
- Scalable and practical for large fleets

Adoption Plan / Next Steps:

- Pilot testing with selected buses
- Evaluate performance